



Molecular dynamics simulations of ion-exchanged glass



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ABSTRACT

Chemically strengthened glass is one of the key enablers of the modern consumer electronics industry. Molecular dynamics simulations offer an opportunity to build enhanced fundamental understanding of the ion exchange process used for chemical strengthening. In this work, we apply molecular modeling techniques to investigate several aspects of ion-exchanged glasses that are either inaccessible through experiment or too difficult to probe through experiments alone. In particular, we use atomistic modeling to study the dependence of the linear network dilation coefficient on the concentration of alkali ions exchanged. We also calculate the evolution of elastic moduli as a result of the ion exchange process. Both of these quantities—the network dilation coefficient and the Young's modulus—are critical parameters for determining the final compressive stress profile of the chemically strengthened glass. Finally, we revisit the issue of structural differences between as-melted and ion-exchanged glasses in terms of local atomic number density around alkali ions. The results of this work point to the unique structures attainable through ion exchange that cannot be achieved through melt-quenching alone.

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1. Introduction

Chemical strengthening of glass via the ion exchange process is a highly effective technique for increasing the practical strength of glass products [1–9]. Compared to other strengthening techniques such as thermal tempering, the ion exchange process offers a significantly higher degree of compressive stress. Ion exchange also enables the strengthening of thin glass sheets and glass products with complicated three-dimensional geometries.

The ion exchange process for chemical strengthening was originally developed by Kistler [6] in 1962. However, it has become a widespread industrial technique for strengthening commercial glass products only more recently. Today, chemically strengthened glass is truly ubiquitous, featured as an integral component of billions of personal electronic devices. One of the keys to this success has been the development of new glass compositions that are specially optimized for the ion exchange process [7–9]. Chemically strengthened glass is also finding its way into new applications such as high-strength substrates for organic electronics [10,11] and as anti-microbial substrates to help prevent the spread of infectious diseases [12].

The ion exchange process is based on immersing an alkali-containing glass in a molten salt bath containing larger alkali ions compared to what are found in the glass itself. During the ion exchange process, the smaller alkali ions from the glass are replaced by larger alkalis from the salt bath. By stuffing these larger alkali ions into sites

previously occupied by the smaller species in the glass, a compressive stress layer can be generated. This compressive stress provides a protective barrier that must be overcome before cracks can propagate in the glass [1,2].

The compressive stress profile generated through the ion exchange process is given by

$$\sigma(z) = -\frac{BE}{1-\nu} [C(z) - C_{\text{avg}}] \quad (1)$$

where E is the Young's modulus of the glass, ν is Poisson's ratio, and $C(z)$ is the local concentration of the substituting alkali ions at a penetration depth z in the glass. The prefactor $E/(1-\nu)$ converts the strain from the ion exchange process into a stress profile. In order for Eq. (1) to satisfy the force balance condition, the average concentration of the substituting cations, C_{avg} , is subtracted from $C(z)$ such that the integral of $\sigma(z)$ over the complete thickness of the glass is zero. In other words, the integrated negative (compressive) stress near the surface of the glass is exactly compensated by the integrated positive (tensile) stress in the interior so that the net force is zero [13–15]. Please note that Eq. (1) does not account for stress relaxation effects, which can act to reduce the magnitude of both compressive and tensile stresses at sufficiently high combinations of salt bath temperature and treatment time [16].

The quantity B in Eq. (1) is the linear network dilation coefficient (LNDC) [15–19], also sometimes called the “Cooper coefficient” in honor of A. R. Cooper, one of the pioneers in understanding the physics of chemically strengthened glass. The concept of the LNDC was originally proposed by Richmond et al. [20] in reference to the

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stresses generated during the nitrogenation of metals. It can be considered as a property analogous to the thermal expansion coefficient of a material. While the thermal expansion coefficient defines the strain of a material per unit change in temperature, the LNDC gives the (linear) strain per unit change in alkali concentration, i.e., the strain per unit of alkali substitution during the ion exchange process [15]. Mathematically, the LNDC can be expressed as

$$B = \frac{1}{3} \left(\frac{1}{V} \frac{\partial V}{\partial C} \right) = \frac{1}{3} \frac{\partial \ln V}{\partial C} \quad (2)$$

where V is the molar volume of the glass.

One of the puzzling features related to the ion exchange process is the anomalous behavior of the LNDC. Specifically, when Eq. (2) is evaluated by measuring the difference in molar volume of as-melted (non-ion-exchanged) glasses between alkali end-member compositions, the value of B is always a factor of 2–4 higher than that deduced from ion-exchanged stress profiles using Eq. (1) together with measured values of E , ν , $\sigma(0)$, $C(0)$, and C_{avg} [2,19]. This so-called “network dilation anomaly” is found even in the limit of short-time salt bath treatments and therefore cannot be explained as an artifact of stress relaxation. From a technological perspective, the lower value of B for ion-exchanged glasses leads to a significantly lower compressive stress than what would be expected based on as-melted versions of the same glass compositions.

In a previous pair of studies [21,22], we performed molecular dynamics simulations of a series of alkali silicate and alkali aluminosilicate glasses to investigate the detailed atomistic origin of the network dilation anomaly. Our simulations showed that the anomalous behavior of the LNDC is a result of different local environments around alkali ions in as-melted versus ion-exchanged versions of the same glass composition. When a larger alkali ion is stuffed into a site previously occupied by a smaller alkali, the glass structure is not able to achieve its fully desired increase in molar volume, such as would occur in an as-melted version of the same composition. These results were confirmed by an independent molecular dynamics study due to Kreski et al. [23]. In our second investigation [22], we performed reverse ion exchange simulations to quantify the level of plasticity in the ion stuffing process. In these reverse ion exchange simulations, the invading alkali ions were swapped back to the original species, as in the as-melted glass. After this reverse exchange process was completed, we found that the original volume of the as-melted glass was fully recovered, indicating that strain induced by ion stuffing is entirely elastic in nature. In other words, the network dilation anomaly is the result of a failure to achieve the fully expected magnitude of elastic strain, and not a result of plasticity in the glass network as has been previously hypothesized [24–28]. Moreover, we showed that the elastic response of the glass to ion stuffing consists of instantaneous and delayed elastic contributions, both of which are fully reversible after substituting the original ions back into the ion-exchanged glass [22].

In the current paper, we extend our molecular dynamics simulations to explore three other features related to the fundamental physics of ion-exchanged glass. First, we consider the dependence of the LNDC on the concentration of alkali ions exchanged, showing that there can be a nonlinear scaling of strain as more alkali ions are exchanged. Second, we investigate the impact of ion exchange on the Young's modulus of the glass. We find that the modulus evolves non-monotonically as a result of the delayed elastic response of the glass following the initial ion substitution. This, of course, has a direct impact on the compressive stress values calculated through Eq. (1). Finally, we revisit the topic of structural differences between as-melted and ion-exchanged glasses of the same composition. In particular, here we calculate the atomic number density of various species around alkali ions to show the intrinsic differences in local environment between alkali sites before and after ion exchange.

2. Simulation procedure

For our molecular dynamics simulations, we consider a series of alkali aluminosilicate glass compositions according to the formula $x\text{Na}_2\text{O} (20 - x)\text{K}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$ (mol %), where $x = 0, 2, 4, \dots, 20$. These compositions have been chosen to cover the full range of alkali concentrations from pure Na_2O to pure K_2O end members. We specifically choose alkali aluminosilicates for this study since the structural and topological environment around aluminum allows for high alkali mobility while maintaining a rigid glass network [29–35]. Hence, alkali aluminosilicate glasses are the preferred choice for commercial chemically strengthened glass products.

The procedure for calculating the as-melted glass structures is similar to that of our previous study [22]. The simulations begin by placing ~20,000 atoms at random positions in a cubic simulation cell with periodic boundary conditions. The initial dimensions of the simulation cell are chosen to match experimental data in literature [36]. The “as-melted” glass structures are obtained through a stepwise quenching starting at high temperature, with combined Metropolis Monte Carlo (MC) [37] and molecular dynamics (MD) [38] that run at each temperature step. The temperature begins at 4200 K and decreases in steps of 300 K down to 900 K. From here, a reduction of 200 K brings the glass to its final temperature of 700 K, which corresponds to a typical laboratory temperature for ion exchange. At each temperature step, the MC run consists of 100 million trial moves under isobaric conditions (i.e., NPT ensemble at atmospheric pressure). The MC runs are performed using our in-house code, Molsym [39–41] with a target acceptance ratio of 0.35 for both individual atomic displacements and cell volume changes. Equilibration during each MC run is determined based on relaxation of the potential energy and volume of the system.

The output structures from the MC runs are used as input for the MD simulations, which are conducted using GROMACS [42]. Each MD run is performed under constant volume (i.e., NVT) conditions for 600 ps using a time step of 1 fs. The output from these MD simulations is then used as input for the next round of simulations at the subsequent temperature. After reaching the final temperature of 700 K, we perform another MD simulation in the NPT ensemble for 8 ns to calculate the final glass structure and molar volume of the system. We have found that a combination of MC and MD runs such as this provides an efficient and accurate procedure for calculating glass structures through atomistic level simulation [43,44]. All of the simulations are performed using the Pedone set of interatomic potentials for silicate systems [45]. Electrostatic interactions are calculated using the Wolf method [46] in the MC simulations and using particle mesh Ewald (PME) [47] in the MD part of the calculation.

After the initial “as-melted” glass is equilibrated at 700 K, the “ion-exchanged” glasses are prepared by a direct substitution of y mol % of Na with K atoms, with $y = 2, 4, 6, 8, 10, 12, 14, 16, 18$, and 20. These substitutions are followed by an MD simulation for 6 ns in the NVT ensemble and an additional 8 ns of equilibration in the NPT ensemble. The temperature is held constant at 700 K since this corresponds to a typical salt bath temperature used during the experimental ion exchange process. The calculation of the LNDC is based on the system's final molar volume before and after the Na exchange with K. Calculated values of molar volume are accurate within roughly $\pm 1\%$. Young's modulus values are subsequently calculated using stress–strain analysis [48] at a temperature of 300 K by straining the glass uniaxially until the point of fracture, allowing the dimensions of the simulation cell parallel and orthogonal to the strain direction to vary independently.

3. Results and discussion

3.1. Concentration dependence of the network dilation coefficient

We first consider the question of whether there is a dependence of the LNDC on the evolving concentration of K^+ ions during the ion

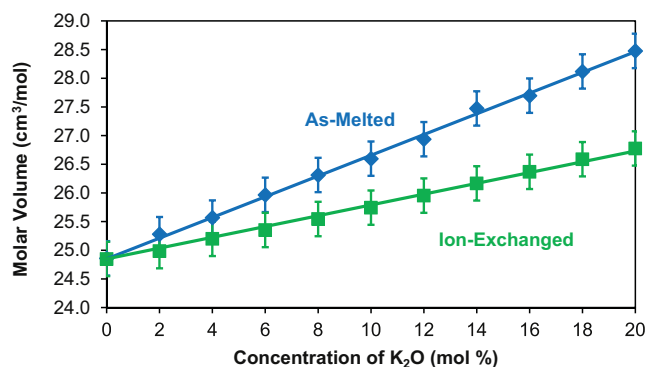


Fig. 1. Molar volume as a function of K_2O concentration in as-melted and ion-exchanged glasses. The as-melted glasses have the composition formula $xNa_2O \cdot (20 - x)K_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (mol %). The ion-exchanged glasses all have a starting composition of $20Na_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (i.e., $x = 20$) and have had different concentrations of Na_2O replaced with K_2O . All molar volumes are calculated at 700 K.

exchange process. In Fig. 1, we plot molar volume as a function of K_2O concentration for as-melted glasses of composition $xNa_2O \cdot (20 - x)K_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (mol %), covering sodium to potassium end members (i.e., x in the range of 0 to 20). In this figure, the ion-exchanged glasses all have an identical starting composition of $20Na_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (i.e., $x = 20$) but have been ion exchanged to different amounts, randomly replacing varying concentrations of Na_2O with K_2O in steps of 2 mol %. As expected, the molar volumes of both the as-melted and ion-exchanged glasses increase with potassium concentration. However, the molar volume of the as-melted glass increases at a rate approximately double that of the ion-exchanged glasses. As with our previous investigations [21,22], this difference in slope is a hallmark of the network dilation anomaly, which results from the inability of the glass to achieve its fully desired elastic strain during ion exchange.

The linear network dilation coefficient can be calculated using Eq. (2) and the molar volume values of the ion-exchanged glasses. As shown in Fig. 2, the LNDC is not a constant but rather increases roughly logarithmically with increasing concentration of alkali ions exchanged. This result indicates that the assumption of a constant B in Eq. (1) may not be strictly valid. The logarithmic increase of B also points to a potential nonlinear advantage when achieving a greater concentration of alkali ions being exchanged. With a greater degree of ion exchange, each of the invading ions is contributing comparatively more strain to the glass owing to this increasing value of B . In other words, a linear increase in $C(z)$ is accompanied by a logarithmic increase in B such that the overall strain shows a slight nonlinearity. In practice, this effect would manifest itself as a deviation from linearity in a molar volume

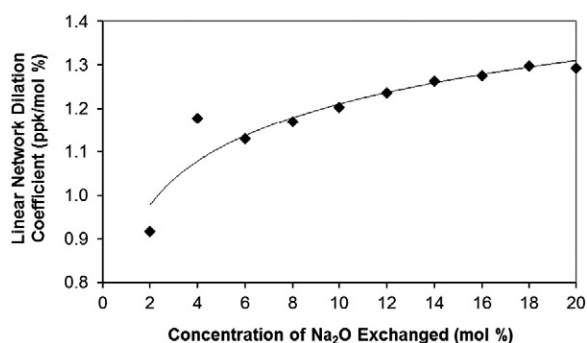


Fig. 2. Linear network dilation coefficient (LNDC) of $20Na_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (mol %) as a function of concentration of Na_2O exchanged. The solid line represents a logarithmic fit to the calculated data points.

versus concentration plot, such as shown in Fig. 1. In the glass compositions under study, the concentration dependence of B is small enough to be considered a second-order effect since the molar volume curve of the ion-exchanged glass in Fig. 1 is approximately linear. This raises the interesting question of what glass compositions might display a more pronounced concentration-dependent dilation coefficient and, therefore, might offer a greater nonlinear advantage in stress profile as governed by Eq. (1).

Interestingly, the logarithmic shape of the LNDC curve in Fig. 2 mirrors that of the composition dependence of the LNDC in our previous work [21,22]. This indicates that there may be some fundamental link between the concentration dependence of B due to partial ion exchange and the composition dependence of B due to the starting fraction of K_2O in the as-melted glass. The main difference is that the magnitude of the effect is smaller in the case of concentration dependence during partial ion exchange.

3.2. Evolution of Young's modulus during ion exchange

Next we turn our attention to the evolution of Young's modulus values during the ion exchange process. In Fig. 3, we show an example stress–strain curve for an as-melted glass with composition $4Na_2O \cdot 16K_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (mol %). The Young's modulus is calculated as the slope of this curve in the initial linear elastic regime (up to a strain of 0.03). At higher values of strain, the elastic response becomes nonlinear before the glass ultimately fractures.

The computed values of the Young's moduli are plotted in Fig. 4 for the as-melted and ion-exchanged glasses. As expected, the modulus values increase with increasing concentration of sodium in the as-melted glass. The modulus values for the ion-exchanged glasses are calculated for a complete ion exchange, i.e., replacing all Na_2O in the as-melted glasses with K_2O , considering two different points in time during the ion exchange process. The red squares plotted in the figure show the modulus values immediately following the initial substitution of all Na with K, but before allowing the structure of the glass to respond to this substitution. At this point in time, the molar volume of the ion-exchanged glass is identical to that of the as-melted glass: the only change is that all of the Na^+ ions have been replaced with K^+ . This initial ion substitution leads to a large increase in modulus values owing to force-fitting of the larger K^+ ions into sites previously occupied by Na^+ .

The second set of data for the ion-exchanged glasses, represented by the green triangles in Fig. 4, show the calculated modulus values after the ion-exchanged glasses are allowed to relax according to the procedure outlined in Section 2. In our previous work [22], we showed that

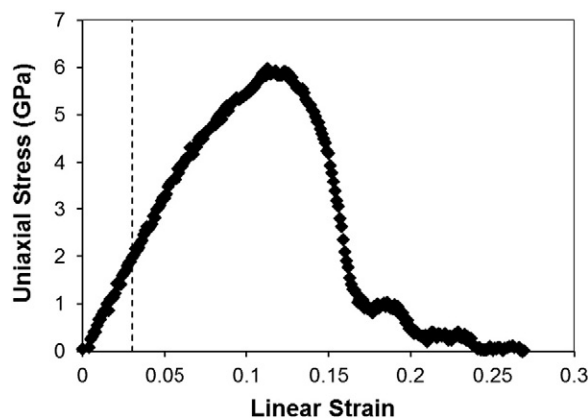


Fig. 3. Example stress–strain curve for the as-melted glass with composition $4Na_2O \cdot 16K_2O \cdot 20Al_2O_3 \cdot 60SiO_2$ (mol %) at a temperature of 300 K. The Young's modulus is computed as the slope of this curve over a range of strains from 0 to 0.03.

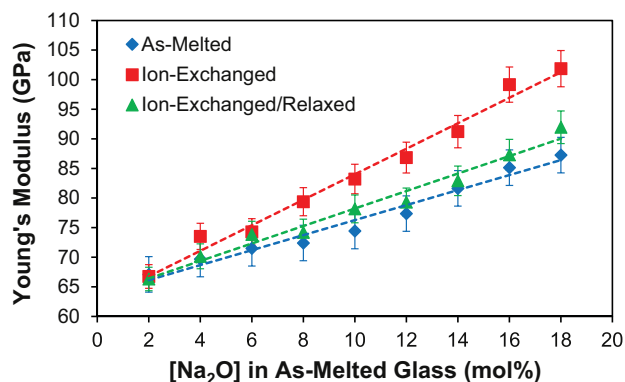


Fig. 4. Calculated Young's moduli for glasses with an as-melted composition of $x\text{Na}_2\text{O} \cdot (20 - x)\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ (mol %) at a temperature of 300 K. Modulus values are shown for the as-melted glasses, the glasses immediately following ion substitution, and after both ion exchange and structural relaxation. Much of the initial increase in modulus values due to ion exchange is relaxed as the volume of the glass expands to reduce localized stresses in the system.

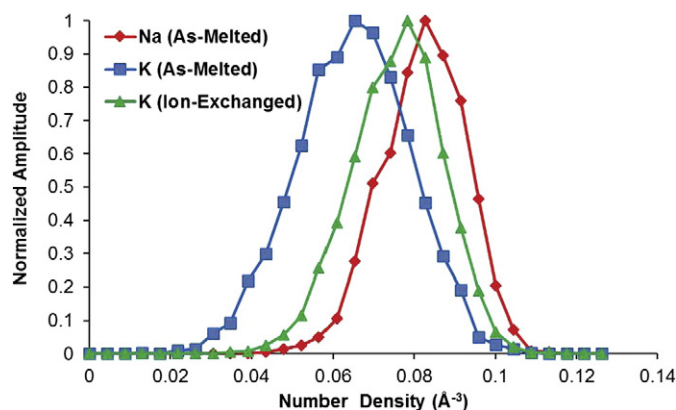


Fig. 5. Number density distribution of all atoms around Na or K within their first-neighbor coordination sphere. The distributions for the as-melted Na and K atoms are calculated based on as-melted $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ or $20\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ compositions, respectively. The number density distribution for the ion-exchanged K is calculated based on the fully ion-exchanged glass with a starting composition of $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$.

this relaxation has a delayed elastic nature. Here, we find that this delayed elastic relaxation involves a decrease in the modulus values as the structure of the glass rearranges to accommodate the larger invading species. Notwithstanding this decrease of the elastic moduli due to relaxation, the final modulus values of the ion-exchanged glass are still greater than those of the initial as-melted glass.

We thus find that the elasticity of the glass follows a nonmonotonic response to ion exchange, with an initial sharp increase as a result of ion substitution and a subsequent decrease due to structural relaxation of the glass. Of course, in experiment these two stages of ion exchange are not neatly separated in time, but rather occur simultaneously in different regions of the glass.

The implication of this result is that the value of Young's modulus in Eq. (1) is not a constant with respect to the penetration depth z . A strictly accurate calculation of the stress profile using Eq. (1) must therefore account for this variation of the Young's modulus through the depth of the glass.

3.3. Atomic number density

Finally, we revisit the subject of the inherent structural differences between as-melted and ion-exchanged glasses. In our prior work [21, 22], we found distinct differences in the local structural environments around potassium in the as-melted and ion-exchanged glasses. Specifically, the invading potassium ions in the ion-exchanged glass exhibit a significantly lower K-O coordination number compared to that found in the original as-melted glass. This is due to the inability of the glass structure to fully relax to achieve the most favorable local environment around the new potassium species. In our simulations, there was no effect of ion exchange on the Q speciation of the network-forming cations: the main difference between as-melted and ion-exchanged structural environments is found near the alkali ions themselves.

Here, we analyze the structural differences between as-melted and ion-exchanged glasses in terms of the differences in total atomic number density around the alkali ions. In all cases, the atomic number density is calculated for neighbors within a radius corresponding to the first minimum of the K-O pair distribution function (3.8 Å). In Fig. 5, we plot the atomic number density distribution of all atoms around Na or K within this first coordination sphere. Here, distributions around the as-melted Na and K atoms are calculated based on the as-melted $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ or $20\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ compositions, respectively. The distribution for the ion-exchanged K is calculated for a fully ion-exchanged glass with a starting composition of

$20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$. Here, we show that the local environment around sodium ions is the most closely packed. While the as-melted potassium ions prefer the most open structure, the potassium ions introduced through ion-exchange are unable to achieve such openness in their environment, instead adopting an intermediate packing density between that of the original as-melted sodium and that of the ideal as-melted potassium. This result is further indication of the structural origin of the lattice dilation anomaly, owing to the inability of the ion-exchanged potassium to achieve its desired local environment such as exhibited in the as-melted glass.

In Fig. 6, we analyze these results in more detail, individually plotting the number density distributions of oxygen, silicon, and aluminum around potassium ions in the same as-melted and ion-exchanged glasses as considered in Fig. 5. In all cases, the neighboring elements are packed more closely to the potassium ions in the ion-exchanged glass compared to the as-melted glass of the same composition.

Next we consider the difference in local atomic number density for the case of a partial ion exchange. In Fig. 7, we plot the number density distributions around alkali in the as-melted $10\text{Na}_2\text{O} \cdot 10\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ (mol %) glass and a partially ion-exchanged version of $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ (mol %), where 10 mol % of the Na_2O has been replaced with K_2O . The trends shown here are consistent with our previous results, with sodium preferring a more tightly packed environment and the ion-exchanged potassium adopting an intermediate structural environment between that of the as-melted sodium and potassium ions. The detailed number densities of oxygen, silicon, and aluminum atoms are plotted in Fig. 8 for these same glasses. Again, the trends are the same regardless of which type of element we consider around the potassium ions.

As we have noted previously [21,22] and has been independently noted by Kreski et al. [23], these intrinsic differences between as-melted and ion-exchanged structures indicate that ion-exchanged glass is a manifestation of the "forbidden glass" concept of Mauro and Loucks [49,50], i.e., a glass structure that is not attainable by cooling from the liquid state using any possible cooling path. This distinct nature of ion-exchanged glass also points to the inadequacy of the fictive temperature concept for describing the nonequilibrium thermodynamic state of the glass [51,52]. In other words, the nonequilibrium state of the ion-exchanged glass cannot be described by any combination of supercooled liquid states, or by any combination of glassy states formed via the standard melt-quenching route. This raises the difficult question of how to describe the nonequilibrium thermodynamics of an ion-exchanged glass. Any successful description must account for both the stress and concentration profiles in the glass, as well as its

complete thermal history. To date, there is no general description of the nonequilibrium thermodynamics of glass that can account for all of these effects.

4. Conclusions

The compressive stress profile of a chemically strengthened glass depends on both the linear network dilation coefficient of the glass, which governs the magnitude of elastic strain associated with ion substitution, and the elastic moduli, which dictate how this strain is converted into stress. The molecular dynamics simulations reported herein indicate that both of these parameters are affected by the details associated with the ion exchange process. In particular, the network dilation coefficient is not necessarily a constant but rather can evolve during the ion exchange process itself. The values of the Young's modulus are also shown to evolve during ion exchange, but displaying a complex nonmonotonic behavior where the initial increase in elastic moduli due to ion substitution is partially relaxed as the glass structure rearranges to accommodate the larger alkali ions. In addition, detailed calculations of the glass structure reveal that the ion-exchanged glass displays structural features unique from those observed in as-melted

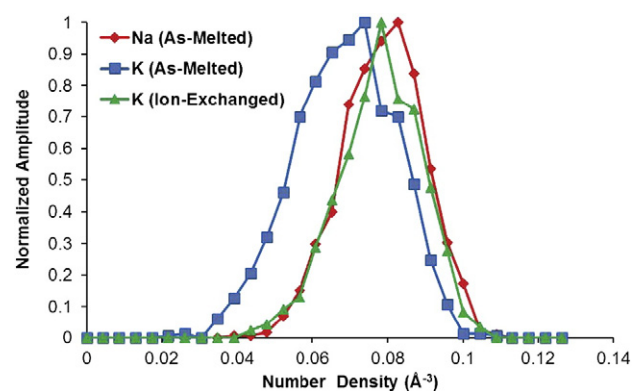


Fig. 7. Number density distribution of all atoms around Na or K within their first-neighbor coordination sphere. Here, the distributions for the as-melted Na and K atoms are calculated from the as-melted $10\text{Na}_2\text{O} \cdot 10\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ (mol %) glass. The number density distribution for the ion-exchanged K is calculated based on a half-ion-exchanged version of $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ (mol %), where 10 mol % of the Na_2O is replaced with K_2O .

glasses of the same composition. These results indicate that ion-exchanged glasses are examples of forbidden glass structures not accessible through variation in thermal history alone.

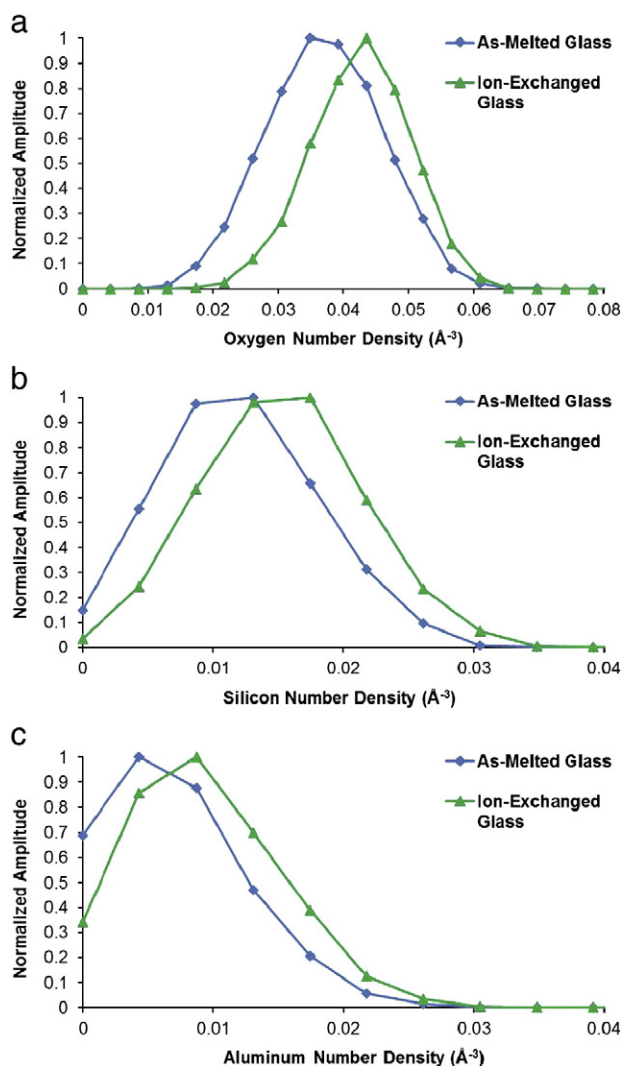


Fig. 6. Number density of (a) oxygen, (b) silicon, and (c) aluminum atoms around potassium in an as-melted glass with composition $20\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ and a fully ion-exchanged glass with a starting composition of $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ (final composition of $20\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$).

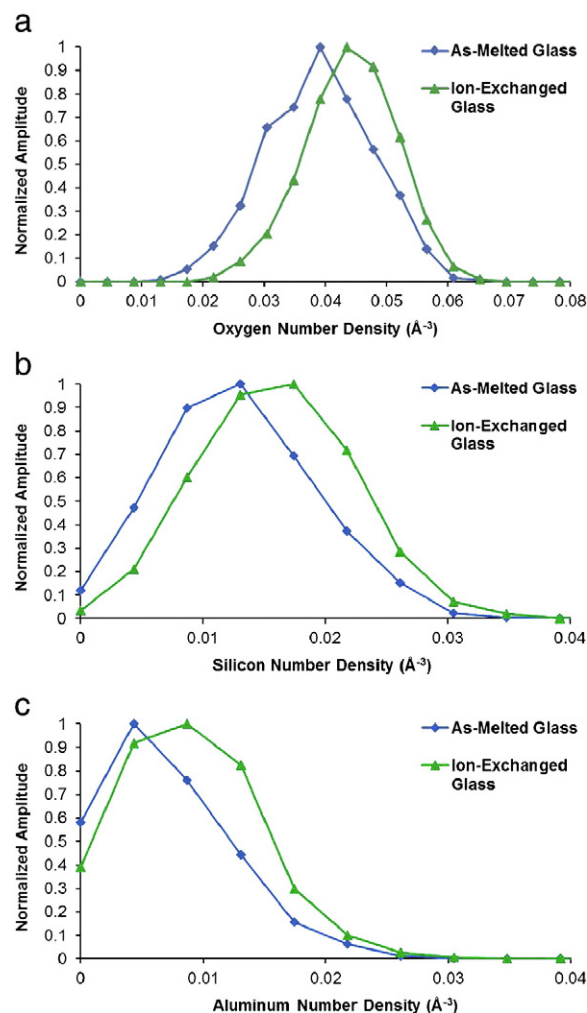


Fig. 8. Number density of (a) oxygen, (b) silicon, and (c) aluminum atoms around potassium in an as-melted glass with composition $10\text{Na}_2\text{O} \cdot 10\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$ and a partially ion-exchanged glass with a starting composition of $20\text{Na}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$, where half of the Na_2O is replaced with K_2O to give a final composition of $10\text{Na}_2\text{O} \cdot 10\text{K}_2\text{O} \cdot 20\text{Al}_2\text{O}_3 \cdot 60\text{SiO}_2$.

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